

A table of Laplace transforms is available for use in questions 1 and 3

1. Find the Laplace transform of

$$f(t) = \int_0^t u^2 - u + e^{-u} du.$$

[10 marks]

2. Calculate the Fourier sine transform,

$$f_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \sin(\omega t) dt$$

of the function $f(t) = \exp(-t)$.

[10 marks]

3. Show that the inverse Laplace transform of

$$\tilde{f}(s) = \frac{5s^2 - 4s - 7}{(s - 3)(s^2 + 4)}$$

can be written in the form $f(t) = ae^{3t} + b \cos 2t + c \sin 2t$, and find the values of the constants a , b and c .

[15 marks]

4. Transform the differential equation

$$\frac{\partial u}{\partial t} + \gamma \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2},$$

where γ and D are constants, to a coordinate system given by $\xi = x - \gamma t$ and $\tau = t$.

[15 marks]

Function	Laplace transform
e^{-at}	$1/(s + a)$
$\sin(\omega t)$	$\omega/(s^2 + \omega^2)$
$\cos(\omega t)$	$s/(s^2 + \omega^2)$
$t^n, n \geq 0$	$n!/s^{n+1}$
$t^p, p > -1$	$\Gamma(p + 1)/s^{p+1}$
$\sinh(at)$	$a/(s^2 - a^2)$
$\cosh(at)$	$s/(s^2 - a^2)$
y'	$s\tilde{y}(s) - y(0)$
y''	$s^2\tilde{y}(s) - sy(0) - y'(0)$